

Semester Project-Spring 2025

CLO 03,04,05

Deadline: 27-04-2025, 11:59PM

Requirements:

Choose a dataset of your own choice from the UCI repository or Kaggle and apply the following:

- 1- KNN
- 2- K Means
- 3- Naïve Bayes Algorithm
- 4- One algorithm of your choice
- 5- Comparison of your selected algorithm with the required algorithms based on results

You may use the following links for this purpose:

- <https://archive.ics.uci.edu/datasets?skip=50&take=25&sort=desc&orderBy=NumHits&se arch=>
- <https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data>

Instructions for the Dataset and Rubrics:

Choose a dataset with a minimum of 4 classes, at least 2.5K instances with at least 20 features. The rubrics are as follows:

1. Dataset Selection - **5 Marks**
2. Classification - **8 Marks**
3. Clustering - **8 Marks**
4. Performance Evaluation (F1 Score, Precision, Recall & Accuracy) - **5 Marks**
5. Visualization - **5 Marks**
6. Comparison and reason of choosing your algorithm - **5 Marks**
7. Efforts to improve the results if the evaluated ones are not satisfactory - **4 Marks**

Note: The Dataset selected for the project should be unique for all the groups across the session 2023 and will be allotted on a first-come, first-served basis.